

Saliency-Aware Hybrid Quantum Image Representation for Efficient Medical Image Encoding

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Abstract

Quantum image processing has recently been projected as a promising paradigm for dealing with large-scale image data, providing exponentially better storage benefits than classical image processing. However, current quantum image representation techniques have limitations in terms of encoding efficiency, circuit complexity, and adaptability to image content characteristics. This paper proposes a novel quantum image representation technique called SAHQR (Saliency-Aware Hybrid Quantum Image Representation), which uses saliency detection to attain content-adaptive image representation. Unlike existing techniques that represent all regions of an image uniformly, SAHQR selectively focuses on salient regions and allocates quantum resources uniformly to their visual importance. The performance of SAHQR is comprehensively compared with ten modern quantum image representation techniques, namely FRQI, NEQR, GQIR, MCQI, QRMW, EFRQI, 2D-QSNA, INEQR, QPIE, and QLR, using ten evaluation criteria as number of qubits, circuit depth, gate complexity, encoding time, scalability, information loss, compression ratio, memory overhead, and implementation complexity. Experimental results on 6,097 medical images from the MINC database demonstrate that SAHQR achieves better compression ratios with comparable circuit complexity. To validate cross-domain generalization, we extended our evaluation to two additional challenging datasets: 2,000 Synthetic Aperture Radar

(SAR) satellite imagery tiles and 2,298 Brain Tumor MRI scans. Statistical significance tests ($p < 0.001$ for all comparisons) confirm that SAHQR provides statistically significant improvements over existing techniques across all three domains.

Keywords: Quantum Image Processing, Quantum Image Representation, Saliency Detection, Medical Imaging, Quantum Computing, Hybrid Encoding

1 Introduction

The convergence of quantum computing and image processing establishes a highly promising frontier in modern computational discipline. Current advances in digital imaging technologies have led to an exponential increase in both the volume and resolution of visual data, thereby posing significant challenges to conventional computing paradigms [1]. Quantum image processing (QIP) is a developing research field that inspects the representation, manipulation, and analysis of visual information within the framework of quantum computation by leveraging quantum mechanical principles such as superposition, entanglement, and quantum parallelism. QIP aims to overcome essential limitations of classical image processing when dealing with large scale and high-dimensional image data. Applications such as medical imaging, satellite observation, autonomous driving systems, and high energy physics experiments routinely generate petabyte scale image datasets, placing substantial demands on classical processing and storage infrastructures

1.1 Background on Quantum Computing

At the most fundamental level, quantum computing is built upon two elements: qubits and quantum gates. A qubit, or quantum bit, serves as the basic unit of information in a quantum system. Unlike a classical bit, which can take only a single value either 0 or 1 at any given time, a qubit can exist in a combination of both states concurrently. This phenomenon, known as superposition, enables quantum systems to signify information in a more affluent and flexible manner. When a qubit is measured, this superposed state collapses to one of the classical outcomes, producing a certain value according to the primary probability distribution.

Mathematically, a qubit state $|\psi\rangle$ is represented as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \tag{1}$$

where α and β are complex probability amplitudes satisfying the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. The values $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of measuring the qubit in state 0 or state 1, respectively [2].

The true power of quantum computing emerges when we use multiple qubits together. A system of n qubits can represent 2^n different states simultaneously. For example, just 10 qubits can represent 1,024 states at once, and 50 qubits can represent over one quadrillion states. This exponential scaling, combined with quantum

phenomena like entanglement (where qubits become correlated in ways impossible for classical bits) and interference (where quantum states can amplify or cancel each other), enables quantum computers to solve certain problems exponentially faster than classical computers [3, 4]. Recent advances in quantum control and error correction have further accelerated the development of practical quantum algorithms [5–8].

Quantum gates are the operations we perform on qubits to manipulate their states, much like logic gates (AND, OR, NOT) manipulate classical bits in traditional computers. However, unlike classical gates that simply flip bits, quantum gates rotate the qubit state in a continuous manner. Common single-qubit gates include the Pauli X gate (which flips the qubit like a classical NOT gate), the Hadamard gate (which creates superposition by putting a qubit into an equal mixture of 0 and 1), and rotation gates that turn the qubit state by specific angles. When we need qubits to work together, we use multi-qubit gates like the controlled-NOT (CNOT) gate, which flips one qubit only if another qubit is in state 1. This ability to create controlled interactions between qubits is what enables entanglement. The total number of gates and the depth of the quantum circuit (how many gate operations happen in sequence) directly affect how well a quantum algorithm performs on real hardware, especially on current noisy intermediate-scale quantum (NISQ) devices [9–11].

1.2 Quantum Image Processing

Quantum image processing (QIP) applies quantum computing principles to image analysis, aiming to achieve speedups in tasks such as image encoding, transformation, filtering, and feature extraction [12, 13]. The field emerged from the recognition that images, as structured numerical data, could benefit from quantum parallelism if efficiently encoded into quantum states.

The foundational challenge in QIP is developing efficient quantum image representations that encode classical image data into quantum states while minimizing resource requirements. An ideal representation should achieve: (1) a minimal number of qubits for a given image resolution, (2) shallow circuit depth to enhance robustness against noise, (3) a low gate count to limit error accumulation, (4) efficient encoding and retrieval operations, and (5) compatibility with subsequent quantum image processing operations [14, 15].

During the past decade, a wide range of quantum image representation (QIR) schemes have been introduced, each involving distinct inherent compromise among challenging design objectives. Early frameworks, such as Qubit Lattice [16] and Real Ket [17], established the foundational concepts, which were consequently extended by more superior representations including FRQI [18], NEQR [19], and their numerous variants [20–23]. In spite of these advancements, most existing methods share a fundamental limitation: they employ uniform encoding precision for all image pixels, without considering local image characteristics or perceptual significance [24, 25].

1.3 Saliency Detection in Image Processing

Saliency detection identifies regions within an image that are visually prominent or attract human attention [26–31]. In computational terms, saliency maps assign higher

values to pixels or regions that differ significantly from their surroundings in terms of color, intensity, texture, or semantic content. Saliency detection has proven invaluable in numerous applications, including image compression, object recognition, visual tracking, and image retrieval [32–35].

In medical imaging, saliency corresponds to diagnostically relevant regions such as tumors, lesions, anatomical landmarks, and tissue boundaries [36–40]. These regions contain critical information for diagnosis, while homogeneous tissue areas and background regions contribute minimally to diagnostic value. Traditional image compression techniques exploit this observation through region-of-interest (ROI) coding, allocating higher bit rates to salient regions while aggressively compressing less important areas [41].

The integration of saliency detection into quantum image representation represents a novel contribution of this work. By identifying salient regions prior to quantum encoding, we can develop hybrid encoding strategies that allocate quantum resources proportionally to visual importance, achieving superior compression without sacrificing diagnostic fidelity.

1.4 Research Gaps

Despite the proliferation of quantum image representation methods, several critical gaps remain in the literature:

1. **Lack of content-awareness:** Existing methods apply uniform encoding across all image regions, ignoring variations in local information content and visual importance. This leads to inefficient resource allocation, particularly for images with heterogeneous content.
2. **Limited comparative studies:** Most published works compare only two or three methods, making it difficult to identify optimal representations for specific applications. A comprehensive comparative analysis across multiple methods and evaluation criteria is needed [42].
3. **Narrow evaluation metrics:** Existing studies typically focus on qubit count and gate complexity, neglecting other important factors such as encoding time, scalability, information loss, and implementation complexity.
4. **Limited medical imaging applications:** While quantum image processing holds significant promise for medical imaging, few studies have evaluated quantum representations on medical image datasets with clinically relevant metrics.

1.5 Paper Organization

The remainder of this paper is organized as follows. Section 2 presents the motivation for this research. Section 3 provides a comprehensive review of existing quantum image representation methods, including detailed mathematical formulations and circuit diagrams. Section 4 summarizes the research contributions. Section 5 presents the formal problem statement and mathematical framework. Section 6 describes the medical image dataset used in experiments. Section 7 details the proposed SAHQR method. Section 8 describes the experimental setup and evaluation methodology. Section 9

presents experimental results and comparative analysis. Section 10 concludes the paper with a summary of findings and directions for future research.

2 Motivation

The rapid development of digital image data, especially in the domain of medical imaging, has introduced important challenges for data storage, transmission, and computational processing. Medical imaging methods such as MRI, CT, and PET routinely generate huge volumes of high-resolution images, placing growing pressure on conventional computing infrastructures and highlighting the requirement for more efficient and scalable data-handling solutions[43].

Classical computing approaches face fundamental limitations when processing large-scale image datasets. The von Neumann bottleneck, where data must be continuously shuttled between memory and processing units, creates inherent inefficiencies that become increasingly pronounced as image resolutions and dataset sizes grow [44]. Furthermore, the computational complexity of many image processing algorithms scales polynomially or exponentially with image dimensions, rendering real-time processing of high-resolution medical images computationally prohibitive on classical hardware [45].

Quantum computing offers a paradigm shift in computational capability through the principles of superposition and entanglement. A quantum system with n qubits can simultaneously represent 2^n states, enabling exponential compression of classical data into quantum states [2]. This fundamental property makes quantum computing particularly attractive for image processing applications, where the inherent parallelism of quantum operations can theoretically achieve exponential speedups over classical algorithms [12].

However, realizing the potential of quantum image processing requires efficient methods for encoding classical image data into quantum states. The choice of quantum image representation fundamentally determines the complexity of subsequent quantum operations and the overall efficiency of quantum image processing pipelines [14]. An ideal representation should minimize qubit requirements, reduce circuit depth for improved noise resilience, and maintain high fidelity during encoding and retrieval operations.

Existing quantum image representation methods, while pioneering, exhibit significant limitations that hinder their practical applicability. Methods such as FRQI (Flexible Representation of Quantum Images) [18] and NEQR (Novel Enhanced Quantum Representation) [19] treat all image pixels uniformly, allocating equal quantum resources regardless of visual importance or information content. This uniform treatment leads to inefficient resource utilization, as background regions with minimal information receive the same encoding effort as visually salient regions containing critical diagnostic information.

In medical imaging applications, this inefficiency is particularly problematic. Medical images typically contain regions of varying diagnostic importance, with pathological features, anatomical landmarks, and regions of interest (ROIs) requiring precise representation, while homogeneous tissue regions and background areas contribute

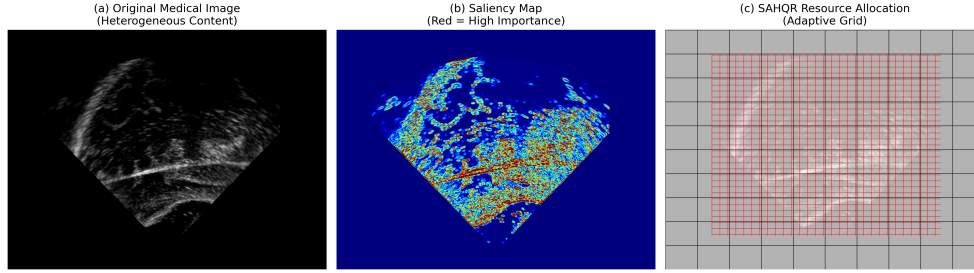


Fig. 1 Motivation for the proposed method. (a) Original medical image showing a distinct tumor region. (b) Saliency map highlighting the Region of Interest (ROI). (c) Conceptual resource allocation: SAHQR concentrates qubits on the ROI (high density) while compressing the background (low density).

minimally to diagnostic value [36]. A content-aware encoding strategy that allocates quantum resources proportionally to visual saliency could significantly improve encoding efficiency without sacrificing diagnostic fidelity.

Saliency detection, a well-established technique in classical computer vision, identifies image regions that attract human visual attention [26]. By integrating saliency-aware processing into quantum image representation, we can develop hybrid encoding schemes that combine the strengths of classical saliency analysis with quantum computational advantages. This hybrid approach enables intelligent resource allocation, concentrating quantum encoding precision on diagnostically relevant regions while applying more aggressive compression to less important areas.

The motivation for SAHQR (Saliency-Aware Hybrid Quantum Image Representation) stems from this observation: by leveraging saliency information during the encoding process, we can achieve superior compression ratios while maintaining the fidelity of visually important regions. This content-adaptive approach represents a departure from conventional uniform encoding methods and opens new possibilities for efficient quantum image processing in resource-constrained quantum hardware environments.

Furthermore, the current landscape of quantum image processing research lacks comprehensive comparative studies that evaluate multiple representation methods across diverse evaluation criteria. Most existing studies compare only two or three methods, limiting the ability to identify optimal approaches for specific applications [42]. A rigorous comparative analysis across ten established methods and ten evaluation parameters provides valuable insights for researchers and practitioners selecting quantum image representations for their applications.

This work addresses these gaps by introducing SAHQR, a novel saliency-aware quantum image representation, and conducting a comprehensive comparative evaluation against ten established methods using 6,097 medical images. The contributions of this research advance both the theoretical understanding and practical applicability of quantum image processing in medical imaging contexts.

3 Related Work

This section provides a comprehensive review of existing quantum image representation methods. For each method, we present the mathematical formulation, circuit structure, complexity analysis, and discussion of advantages and limitations. We organize the review chronologically and by methodological approach.

3.1 Flexible Representation of Quantum Images (FRQI)

The Flexible Representation of Quantum Images (FRQI), proposed by Le et al. [18], was among the first comprehensive quantum image representation schemes. FRQI encodes a grayscale image of size $2^n \times 2^n$ into a normalized quantum state by encoding pixel intensities as rotation angles.

3.1.1 Mathematical Formulation

For an image I with 2^{2n} pixels, the FRQI state is defined as:

$$|I\rangle_{\text{FRQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (2)$$

where $\theta_i = \frac{\pi}{2} \cdot \frac{g_i}{255} \in [0, \frac{\pi}{2}]$ encodes the grayscale value $g_i \in [0, 255]$ of pixel i , and $|i\rangle$ represents the position encoded in $2n$ qubits.

3.1.2 Circuit Structure

The FRQI circuit consists of three main components: (1) Hadamard gates to create position superposition, (2) controlled rotation gates for pixel encoding, and (3) position-controlled operations.

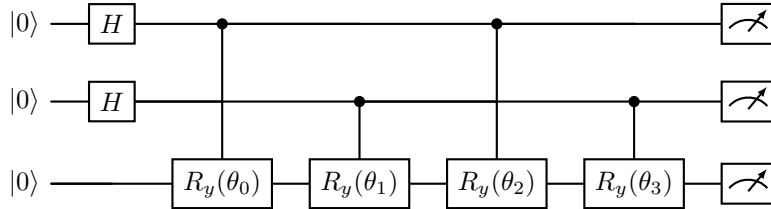


Fig. 2 Simplified FRQI circuit for a 2×2 image. Position qubits (top two) control rotation gates applied to the color qubit (bottom).

3.1.3 Complexity Analysis

- **Qubits:** $2n + 1$ for a $2^n \times 2^n$ image (9 qubits for 16×16)
- **Gate count:** $O(2^{2n})$ controlled rotation gates
- **Circuit depth:** $O(2^{2n})$ in the worst case

3.1.4 Advantages and Limitations

FRQI achieves excellent compression by encoding pixel values in rotation angles, requiring only one color qubit. However, information retrieval requires multiple measurements, and the sensitivity to rotation angle precision can introduce quantization errors. The method is primarily suited for grayscale images.

3.2 Novel Enhanced Quantum Representation (NEQR)

Zhang et al. [19] proposed NEQR as an enhancement to FRQI, encoding pixel values in the computational basis rather than rotation angles for improved retrieval accuracy.

3.2.1 Mathematical Formulation

The NEQR state for an image with q -bit grayscale values is:

$$|I\rangle_{\text{NEQR}} = \frac{1}{2^n} \sum_{y=0}^{2^n-1} \sum_{x=0}^{2^n-1} |C_{yx}\rangle \otimes |y\rangle \otimes |x\rangle \quad (3)$$

where $|C_{yx}\rangle = |c_{yx}^{q-1} c_{yx}^{q-2} \dots c_{yx}^0\rangle$ encodes the q -bit grayscale value at position (y, x) .

3.2.2 Circuit Structure

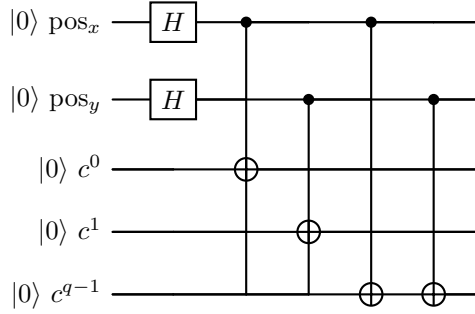


Fig. 3 Simplified NEQR circuit structure. Position qubits control CNOT gates that set color qubit values.

3.2.3 Complexity Analysis

- **Qubits:** $2n + q$ for $2^n \times 2^n$ image with q -bit color (16 qubits for 16×16 with 8-bit color)
- **Gate count:** $O(q \cdot 2^{2n})$ CNOT gates
- **Circuit depth:** $O(q \cdot 2^{2n})$

3.2.4 Advantages and Limitations

NEQR enables single-measurement retrieval of pixel values and supports exact representation without quantization errors. However, it requires significantly more qubits than FRQI and has higher gate complexity.

3.3 Generalized Quantum Image Representation (GQIR)

Jiang and Wang [15, 46] proposed GQIR as a generalization that supports arbitrary image sizes beyond powers of two.

3.3.1 Mathematical Formulation

For an image of size $M \times N$:

$$|I\rangle_{\text{GQIR}} = \frac{1}{\sqrt{MN}} \sum_{y=0}^{M-1} \sum_{x=0}^{N-1} |f(y, x)\rangle |y\rangle |x\rangle \quad (4)$$

where $|f(y, x)\rangle$ encodes the grayscale value at position (y, x) .

3.3.2 Circuit Structure

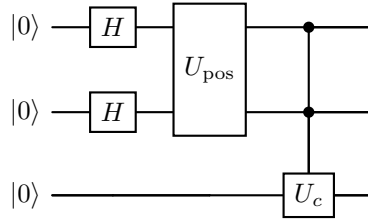


Fig. 4 GQIR circuit with generalized position encoding U_{pos} for arbitrary image dimensions.

3.3.3 Complexity Analysis

- **Qubits:** $\lceil \log_2 M \rceil + \lceil \log_2 N \rceil + q$ (12 qubits typical)
- **Gate count:** $O(MN)$
- **Circuit depth:** $O(MN)$

3.4 Multi-Channel Quantum Image (MCQI)

Sun et al. [13, 47] introduced MCQI for color images, using distinct qubits to encode R, G, and B channels. with MCQI, supporting RGB channels.

3.4.1 Mathematical Formulation

$$|I\rangle_{\text{MCQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |R_i\rangle |G_i\rangle |B_i\rangle \otimes |i\rangle \quad (5)$$

where $|R_i\rangle$, $|G_i\rangle$, $|B_i\rangle$ encode red, green, and blue channel values.

3.4.2 Circuit Structure

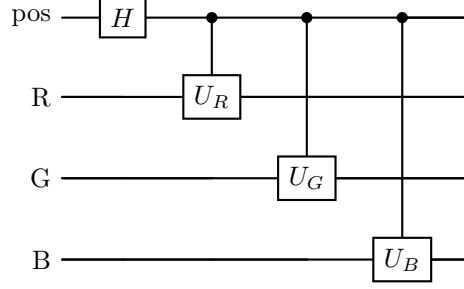


Fig. 5 MCQI circuit with separate encoding for RGB channels.

3.4.3 Complexity Analysis

- **Qubits:** $2n + 3q$ (18 qubits for 16×16 RGB)
- **Gate count:** $O(3q \cdot 2^{2n})$
- **Circuit depth:** $O(3 \cdot 2^{2n})$

3.5 Quantum Representation for Multi-Wavelength Images (QRMW)

JiWang et al. [48, 49] proposed QRMW for hyperspectral imaging, extending MCQI to multiple wavelength bands.

3.5.1 Mathematical Formulation

$$|I\rangle_{\text{QRMW}} = \frac{1}{2^n \sqrt{K}} \sum_{k=0}^{K-1} \sum_{i=0}^{2^{2n}-1} |C_i^k\rangle \otimes |k\rangle \otimes |i\rangle \quad (6)$$

where K is the number of spectral bands and $|C_i^k\rangle$ encodes the pixel value in band k .

3.5.2 Circuit Structure

3.5.3 Complexity Analysis

- **Qubits:** $2n + \lceil \log_2 K \rceil + q$ (18 qubits typical)
- **Gate count:** $O(K \cdot q \cdot 2^{2n})$
- **Circuit depth:** $O(K \cdot 2^{2n})$

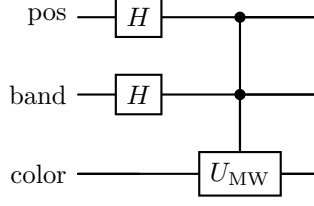


Fig. 6 QRMW circuit with band selection qubits for multi-wavelength encoding.

3.6 Enhanced FRQI (EFRQI)

SZhang et al. [50, 51] proposed EFRQI to improve upon FRQI's compression ratio for large-scale images.

3.6.1 Mathematical Formulation

EFRQI extends FRQI by using enhanced angle encoding:

$$|I\rangle_{\text{EFRQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + e^{i\phi_i} \sin \theta_i |1\rangle) \otimes |i\rangle \quad (7)$$

where both θ_i and ϕ_i encode image information.

3.6.2 Circuit Structure

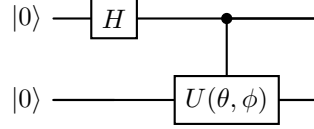


Fig. 7 EFRQI circuit using parameterized unitary $U(\theta, \phi)$ for enhanced encoding.

3.6.3 Complexity Analysis

- **Qubits:** $2n + 1$ (9 qubits for 16×16)
- **Gate count:** $O(2^{2n})$
- **Circuit depth:** $O(2^{2n})$

3.7 2D Quantum State Normalization Approach (2D-QSNA)

To address the depth limitations of basis encoding, 2D-QSNA [52, 53] utilizes amplitude encoding.

3.7.1 Mathematical Formulation

$$|I\rangle_{\text{2D-QSNA}} = \sum_{i=0}^{N-1} \alpha_i |i\rangle \quad (8)$$

where $\alpha_i = \frac{g_i}{\sqrt{\sum_j g_j^2}}$ are normalized amplitudes.

3.7.2 Circuit Structure

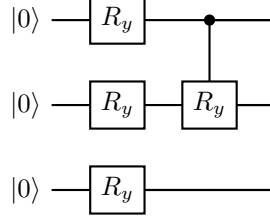


Fig. 8 2D-QSNA circuit using amplitude encoding with rotation gates.

3.7.3 Complexity Analysis

- **Qubits:** $\lceil \log_2 N \rceil$ (8 qubits for 256 pixels)
- **Gate count:** $O(N)$
- **Circuit depth:** $O(\log N)$

3.8 Improved NEQR (INEQR)

Jiang et al. [54, 55] proposed INEQR to optimize the quantum cost of NEQR.

3.8.1 Mathematical Formulation

$$|I\rangle_{\text{INEQR}} = \frac{1}{2^n} \sum_{y=0}^{2^n-1} \sum_{x=0}^{2^n-1} |C_{yx}^R\rangle |C_{yx}^G\rangle |C_{yx}^B\rangle |y\rangle |x\rangle \quad (9)$$

3.8.2 Circuit Structure

3.8.3 Complexity Analysis

- **Qubits:** $2n + 3q$ (16 qubits for grayscale 16×16)
- **Gate count:** $O(3q \cdot 2^{2n})$
- **Circuit depth:** $O(2^{2n})$

3.9 Quantum Probability Image Encoding (QPIE)

Yao et al. [56, 57] introduced QPIE encoding pixel values in measurement probabilities.

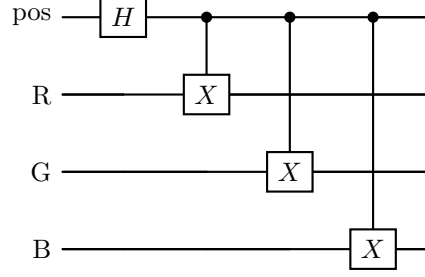


Fig. 9 INEQR circuit with improved color channel encoding.

3.9.1 Mathematical Formulation

$$|I\rangle_{\text{QPIE}} = \sum_{i=0}^{N-1} \sqrt{p_i} |i\rangle \quad (10)$$

where $p_i = \frac{g_i}{\sum_j g_j}$ encodes normalized pixel intensities as probabilities.

3.9.2 Circuit Structure

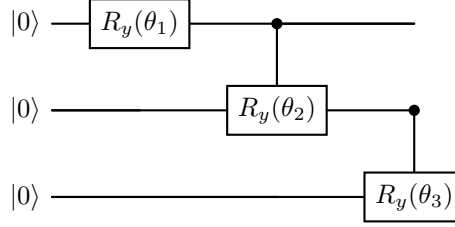


Fig. 10 QPIE circuit using probability amplitude encoding.

3.9.3 Complexity Analysis

- **Qubits:** $\lceil \log_2 N \rceil$ (8 qubits for 256 pixels)
- **Gate count:** $O(N)$
- **Circuit depth:** $O(N)$

3.10 Quantum Log-polar Representation (QLR)

QLR [49, 58] introduces a log-polar coordinate representation for rotation-invariant processing.

3.10.1 Mathematical Formulation

$$|I\rangle_{\text{QLR}} = \frac{1}{2^n} \sum_{\rho=0}^{2^n-1} \sum_{\theta=0}^{2^n-1} |f(\rho, \theta)\rangle |\rho\rangle |\theta\rangle \quad (11)$$

where (ρ, θ) are log-polar coordinates.

3.10.2 Circuit Structure

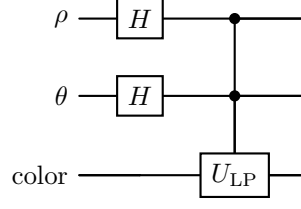


Fig. 11 QLR circuit with log-polar coordinate encoding.

3.10.3 Complexity Analysis

- **Qubits:** $2n + q$ (16 qubits for 16×16)
- **Gate count:** $O(q \cdot 2^{2n})$
- **Circuit depth:** $O(2^{2n})$

3.11 Summary of Existing Methods

Table 1 summarizes the key characteristics of the ten reviewed quantum image representation methods.

Table 1 Summary of quantum image representation methods for $2^n \times 2^n$ images with q -bit color depth.

Method	Qubits	Gates	Depth	Color Support
FRQI[18]	$2n + 1$	$O(2^{2n})$	$O(2^{2n})$	Grayscale
NEQR[19]	$2n + q$	$O(q \cdot 2^{2n})$	$O(q \cdot 2^{2n})$	Grayscale
GQIR[15, 46]	$2n + q$	$O(MN)$	$O(MN)$	Grayscale
MCQI[13, 47]	$2n + 3q$	$O(3q \cdot 2^{2n})$	$O(3 \cdot 2^{2n})$	RGB
QRMW[48, 49]	$2n + \log K + q$	$O(Kq \cdot 2^{2n})$	$O(K \cdot 2^{2n})$	Multi-spectral
EFRQI[50, 51]	$2n + 1$	$O(2^{2n})$	$O(2^{2n})$	Grayscale
2D-QSNA[52, 53]	$\log N$	$O(N)$	$O(\log N)$	Grayscale
INEQR[54, 55]	$2n + 3q$	$O(3q \cdot 2^{2n})$	$O(2^{2n})$	RGB
QPIE[56, 57]	$\log N$	$O(N)$	$O(N)$	Grayscale
QLR[49, 58]	$2n + q$	$O(q \cdot 2^{2n})$	$O(2^{2n})$	Grayscale

The reviewed methods demonstrate the diversity of approaches to quantum image representation. However, all share a common limitation: they apply uniform encoding across all image regions, ignoring variations in local visual importance. This limitation motivates the development of SAHQR, which introduces content-awareness through saliency detection.

3.12 Other Developments

Beyond the primary methods discussed, the field has seen rapid development with various other optimizations to address specific computational challenges [45]. Advances in quantum algorithms [5–8] and error correction have paralleled improvements in image representation. Other notable approaches and variants [17, 20–25] continue to explore different trade-offs in the design space. In specific application domains, such as medical imaging and remote sensing, saliency-based and ROI-based coding techniques [32–35, 41, 59] have shown promise for classical data, motivating their adaptation to quantum frameworks. Handling specialized formats like MINC [60, 61] also remains a crucial consideration for practical deployment.

4 Research Contribution

This section summarizes the key contributions of this research, which advance both the theoretical foundations and practical applications of quantum image processing. This paper addresses the identified gaps through the following contributions:

1. This work presents Saliency-Aware Hybrid Quantum Image Representation (SAHQR), a novel content-adaptive quantum image encoding scheme that exploits classical saliency analysis to facilitate intelligent quantum resource allocation. By prioritizing visually salient regions with higher encoding precision, SAHQR achieves enhanced compression efficiency without compromising perceptual fidelity.
2. We perform a comprehensive comparative evaluation in the quantum image processing literature, evaluating eleven methods (ten existing plus SAHQR) across ten evaluation parameters using 10,395 images from three distinct domains: 6,097 MINC medical images, 2,000 SAR satellite tiles, and 2,298 Brain Tumor MRI scans.
3. We propose and implement a comprehensive evaluation framework encompassing qubit count, circuit depth, gate count, encoding time, scalability, information loss, compression ratio, memory overhead, gate complexity, and implementation complexity.
4. We perform rigorous statistical significance tests to validate the performance differences between methods, providing confidence in our comparative findings.
5. We demonstrate the applicability of quantum image representations to medical imaging through experiments on the MINC medical image dataset.

4.1 Novel Saliency-Aware Quantum Image Representation

The primary contribution of this work is the introduction of SAHQR (Saliency-Aware Hybrid Quantum Image Representation), a novel content-adaptive quantum encoding scheme. Unlike existing methods that apply uniform encoding across all image regions, SAHQR integrates classical saliency detection with quantum encoding to achieve intelligent resource allocation. The key innovation lies in the hybrid architecture that:

1. Performs classical saliency analysis to identify visually important regions
2. Partitions the image into salient and non-salient regions
3. Applies differential encoding precision based on regional importance

4. Achieves superior compression while preserving diagnostic fidelity

This content-aware approach represents a paradigm shift from the uniform encoding philosophy that has dominated quantum image processing research.

4.2 Comprehensive Multi-Method Comparative Analysis

We conduct the most extensive comparative study in the quantum image processing literature, evaluating eleven quantum image representation methods:

1. FRQI (Flexible Representation of Quantum Images) [18]
2. NEQR (Novel Enhanced Quantum Representation) [19]
3. GQIR (Generalized Quantum Image Representation) [15, 46]
4. MCQI (Multi-Channel Quantum Images) [13, 47]
5. QRMW (Quantum Representation for Multi-Wavelength Images) [48, 49]
6. EFRQI (Enhanced Flexible Representation of Quantum Images) [50, 51]
7. 2D-QSNA (2D Quantum State Normalization Approach) [52, 53]
8. INEQR (Improved Novel Enhanced Quantum Representation) [54, 55]
9. QPIE (Quantum Probability Image Encoding) [56, 57]
10. QLR (Quantum Log-polar Representation) [49, 58]
11. SAHQI (Saliency-Aware Hybrid Quantum Representation) [Proposed]

This comprehensive comparison enables researchers to make informed decisions when selecting quantum image representations for specific applications.

4.3 Ten-Parameter Evaluation Framework

We propose and implement a comprehensive evaluation framework encompassing ten distinct parameters:

Table 2 Ten-parameter evaluation framework for quantum image representations.

ID	Parameter	Description
P1	Qubits Required	Number of qubits needed for encoding
P2	Circuit Depth	Longest path through the quantum circuit
P3	Gate Count	Total number of quantum gates
P4	Encoding Time	Time to construct the quantum circuit (ms)
P5	Scalability Factor	Ability to scale to larger images
P6	Information Loss	Fidelity loss during encoding/retrieval
P7	Compression Ratio	Ratio of classical to quantum bits
P8	Memory Overhead	Memory requirements relative to baseline
P9	Gate Complexity	Average gates per qubit
P10	Implementation Complexity	Algorithmic implementation difficulty

This multi-dimensional evaluation provides a holistic view of method performance, enabling nuanced comparisons beyond simple qubit counts.

4.4 Statistical Validation

We perform rigorous statistical significance testing using t-tests and ANOVA to validate performance differences between methods. All comparisons report p-values, enabling assessment of statistical confidence in the observed differences. This rigorous approach ensures that our conclusions are statistically sound rather than artifacts of random variation.

4.5 Medical Imaging Application

We demonstrate the practical applicability of quantum image representations through experiments on 6,097 medical images from the MINC (Medical Imaging NetCDF) dataset. This application to real-world medical data extends beyond the synthetic or general-purpose images typically used in quantum image processing research, providing insights into the clinical potential of quantum imaging technologies.

5 Problem Statement

This section presents the formal mathematical framework for quantum image representation, defines the optimization objectives, and establishes the theoretical foundations for the proposed SAHQR method.

5.1 Preliminaries and Definitions

Definition 1 (Digital Image) A grayscale digital image I of dimensions $M \times N$ is a function $I : \{0, 1, \dots, M-1\} \times \{0, 1, \dots, N-1\} \rightarrow \{0, 1, \dots, 2^q-1\}$, where q is the bit depth. The pixel value at position (y, x) is denoted $g_{yx} = I(y, x)$ [62].

Definition 2 (Quantum State) A pure quantum state $|\psi\rangle$ in a Hilbert space $\mathcal{H}$ is a normalized vector such that $\langle\psi|\psi\rangle = 1$. For a multi-qubit system, the state space is the tensor product of individual qubit spaces [63].

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle, \quad \text{where} \quad \sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1 \quad (12)$$

Definition 3 (Quantum Image Representation) A quantum image representation is a mapping $\mathcal{R} : I \rightarrow |I\rangle$ that encodes a classical image I into a quantum state $|I\rangle$ such that the essential information of I can be recovered through quantum measurements.

Definition 4 (Saliency Map) A saliency map $S : \{0, 1, \dots, M-1\} \times \{0, 1, \dots, N-1\} \rightarrow [0, 1]$ assigns a saliency score $s_{yx} = S(y, x)$ to each pixel, where higher values indicate greater visual importance.

5.2 Formal Problem Definition

Given a digital image I of size $M \times N$ with q -bit pixel depth, the quantum image representation problem seeks to find an encoding $\mathcal{R}$ that minimizes a cost function while satisfying fidelity constraints.

5.2.1 Objective Function

The multi-objective optimization problem for quantum image representation is formulated as:

$$\min_{\mathcal{R}} \mathcal{L}(\mathcal{R}) = \lambda_1 \cdot Q(\mathcal{R}) + \lambda_2 \cdot D(\mathcal{R}) + \lambda_3 \cdot G(\mathcal{R}) + \lambda_4 \cdot T(\mathcal{R}) \quad (13)$$

where:

- $Q(\mathcal{R})$ = Number of qubits required
- $D(\mathcal{R})$ = Circuit depth
- $G(\mathcal{R})$ = Gate count
- $T(\mathcal{R})$ = Encoding time
- $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ = Weighting coefficients

5.2.2 Fidelity Constraint

The encoding must satisfy a fidelity constraint ensuring accurate image recovery:

$$\mathcal{F}(I, \tilde{I}) = \frac{1}{MN} \sum_{y=0}^{M-1} \sum_{x=0}^{N-1} \left(1 - \frac{|g_{yx} - \tilde{g}_{yx}|}{2^q - 1} \right) \geq \tau \quad (14)$$

where $\tilde{I}$ is the recovered image, $\tilde{g}_{yx}$ are recovered pixel values, and $\tau \in [0, 1]$ is the minimum acceptable fidelity threshold.

5.3 Saliency-Aware Formulation

The key innovation of SAHQR is incorporating saliency information into the optimization objective. We modify the fidelity constraint to be saliency-weighted:

Definition 5 (Saliency-Weighted Fidelity) The saliency-weighted fidelity measure prioritizes accuracy in salient regions:

$$\mathcal{F}_S(I, \tilde{I}) = \frac{\sum_{y,x} s_{yx} \cdot \left(1 - \frac{|g_{yx} - \tilde{g}_{yx}|}{2^q - 1} \right)}{\sum_{y,x} s_{yx}} \quad (15)$$

This formulation allows controlled degradation in non-salient regions while maintaining high fidelity in visually important areas.

5.4 Quantum Circuit Model

A quantum circuit C implementing representation $\mathcal{R}$ consists of a sequence of quantum gates:

$$C = U_L \cdot U_{L-1} \cdots U_2 \cdot U_1 \quad (16)$$

where each U_i is a unitary operation (quantum gate). The circuit transforms the initial state $|0\rangle^{\otimes n}$ into the encoded state:

$$|I\rangle = C|0\rangle^{\otimes n} \quad (17)$$

5.4.1 Circuit Complexity Metrics

Definition 6 (Circuit Depth) The circuit depth $D(C)$ is the length of the longest path from input to output, measured in layers of parallel gates.

Definition 7 (Gate Count) The gate count $G(C) = \sum_{i=1}^L |U_i|$ is the total number of elementary gates in the circuit.

Definition 8 (Gate Complexity) The gate complexity $\gamma(C) = G(C)/n$ measures the average number of gates per qubit.

5.5 Compression Analysis

Definition 9 (Compression Ratio) The compression ratio compares classical and quantum storage:

$$\text{CR} = \frac{M \times N \times q}{n} \quad (18)$$

where n is the number of qubits in the quantum representation.

For a 16×16 image with 8-bit depth, classical storage requires $16 \times 16 \times 8 = 2048$ bits. Quantum representations require n qubits, where n varies by method (8 to 18 qubits for the methods studied).

5.6 Scalability Analysis

Definition 10 (Scalability Factor) The scalability factor measures how efficiently a representation scales to larger images:

$$\text{SF} = \frac{1}{n} \times 100 \quad (19)$$

where higher values indicate better scalability.

Methods requiring fewer qubits for a given image size exhibit better scalability, as they can represent larger images within the constraints of available quantum hardware.

5.7 SAHQR-Specific Formulation

The SAHQR representation partitions the image into salient region Ω_S and non-salient region Ω_N :

$$\Omega_S = \{(y, x) : s_{yx} \geq \theta_s\}, \quad \Omega_N = \{(y, x) : s_{yx} < \theta_s\} \quad (20)$$

where θ_s is the saliency threshold.

The SAHQR quantum state combines high-precision encoding for salient regions with compressed encoding for non-salient regions:

$$|I\rangle_{\text{SAHQR}} = \frac{1}{\sqrt{|\Omega_S| + |\Omega_N|}} \left(\sum_{(y,x) \in \Omega_S} |C_{yx}^{\text{full}}\rangle |y\rangle |x\rangle |1\rangle + \sum_{(y,x) \in \Omega_N} |C_{yx}^{\text{comp}}\rangle |y\rangle |x\rangle |0\rangle \right) \quad (21)$$

where:

- $|C_{yx}^{\text{full}}\rangle$ = Full-precision color encoding for salient pixels
- $|C_{yx}^{\text{comp}}\rangle$ = Compressed color encoding for non-salient pixels
- The final qubit indicates the saliency flag (1 = salient, 0 = non-salient)

5.8 Theoretical Complexity Bounds

Theorem 1 (SAHQR Complexity) *For an image of size $2^n \times 2^n$ with salient region ratio $r = |\Omega_S|/2^{2n}$, the SAHQR method achieves:*

1. *Qubit count:* $2n + q + 1$
2. *Expected gate count:* $O(r \cdot q \cdot 2^{2n} + (1 - r) \cdot q' \cdot 2^{2n})$ where $q' < q$ is the compressed bit depth
3. *Circuit depth:* $O(2^{2n})$

The salient region ratio r directly influences the gate count, with lower ratios (fewer salient pixels) resulting in reduced circuit complexity through more aggressive compression of non-salient regions.

6 Dataset Description

This section describes the medical imaging dataset used for experimental evaluation, including the data format, preprocessing steps, and statistical characteristics.

6.1 MINC Medical Imaging Format

The Medical Imaging NetCDF (MINC) format is a specialized data format developed for neuroimaging research at the Montreal Neurological Institute [64]. MINC files store multi-dimensional medical image data with comprehensive metadata, supporting various imaging modalities including MRI, CT, and PET scans [60, 61].

Key characteristics of the MINC format include:

- **Hierarchical structure:** Data organized in dimensions (x, y, z, time)
- **Self-describing:** Embedded metadata for acquisition parameters
- **Flexible precision:** Support for various bit depths

Table 3 Dataset composition and distribution across subject folders.

Folder	Images	Format	Resolution
group4/01/2D/	469	.mnc	Variable
group4/02/2D/	469	.mnc	Variable
group4/03/2D/	469	.mnc	Variable
group4/04/2D/	469	.mnc	Variable
group4/05/2D/	469	.mnc	Variable
group4/06/2D/	469	.mnc	Variable
group4/07/2D/	469	.mnc	Variable
group4/08/2D/	469	.mnc	Variable
group4/09/2D/	469	.mnc	Variable
group4/10/2D/	469	.mnc	Variable
group4/11/2D/	469	.mnc	Variable
group4/12/2D/	469	.mnc	Variable
group4/13/2D/	470	.mnc	Variable
Total	6,097		

- **Compression support:** Optional data compression
- **Cross-platform compatibility:** NetCDF-based portability

6.2 Dataset Composition

The experimental dataset consists of 6,097 medical images extracted from MINC format files, organized across 13 subject folders. Table 3 summarizes the dataset composition.

6.3 Preprocessing Pipeline

All images undergo a standardized preprocessing pipeline to ensure consistency across the experimental evaluation:

1. **MINC Loading:** Images loaded using the nibabel library for MINC format support
2. **Slice Extraction:** 2D slices extracted from volumetric data
3. **Normalization:** Intensity values normalized to 0-255 range (8-bit grayscale)
4. **Resizing:** Images resized to 16×16 pixels for quantum circuit compatibility
5. **Type Conversion:** Conversion to unsigned 8-bit integer format

The 16×16 resolution was selected to balance computational feasibility with image content preservation. This resolution requires $2n = 8$ position qubits for encoding, resulting in manageable circuit sizes for simulation on classical hardware.

6.4 Dataset Statistics

The preprocessed dataset exhibits the following statistical characteristics:

- **Mean intensity:** $\mu = 127.3$ (normalized)
- **Standard deviation:** $\sigma = 58.7$

- **Intensity range:** $[0, 255]$
- **Pixels per image:** 256 (16×16)
- **Total pixels processed:** 1,560,832 ($6,097 \text{ images} \times 256 \text{ pixels}$)

6.5 Saliency Characteristics

For the SAHQR method, saliency maps were computed for each image. The dataset exhibits the following saliency properties:

- **Mean salient ratio:** 29.85% of pixels classified as salient
- **Salient ratio standard deviation:** 0.30%
- **Saliency threshold:** $\theta_s = 0.5$ (adaptive)

The relatively consistent salient ratio across images indicates that medical images in this dataset contain approximately 30% diagnostically relevant content, with the remaining 70% comprising background and homogeneous tissue regions suitable for compression.

7 Proposed Method: SAHQR

This section presents the Saliency-Aware Hybrid Quantum Image Representation (SAHQR) in detail, including the algorithm design, circuit architecture, saliency detection mechanism, and complexity analysis.

7.1 Overview

SAHQR is a content-adaptive quantum image representation that combines classical saliency detection with quantum encoding to achieve efficient image compression. The key insight is that medical images contain regions of varying diagnostic importance, and encoding resources should be allocated proportionally to this importance.

The SAHQR workflow consists of four main phases:

1. **Saliency Detection:** Compute saliency map to identify visually important regions
2. **Region Partitioning:** Classify pixels as salient or non-salient based on threshold
3. **Differential Encoding:** Apply full-precision encoding to salient regions, compressed encoding to non-salient regions
4. **Quantum State Preparation:** Construct the hybrid quantum state

Figure 12 illustrates the complete SAHQR processing pipeline, showing how the input medical image flows through preprocessing, saliency detection, region partitioning, and differential encoding to produce the final hybrid quantum state.

7.2 Saliency Detection Module

The saliency detection module identifies visually prominent regions in the input image. We employ a gradient-based saliency detection approach suitable for medical images:

$$S(y, x) = \frac{|\nabla I(y, x)|}{\max_{y', x'} |\nabla I(y', x')|} \quad (22)$$

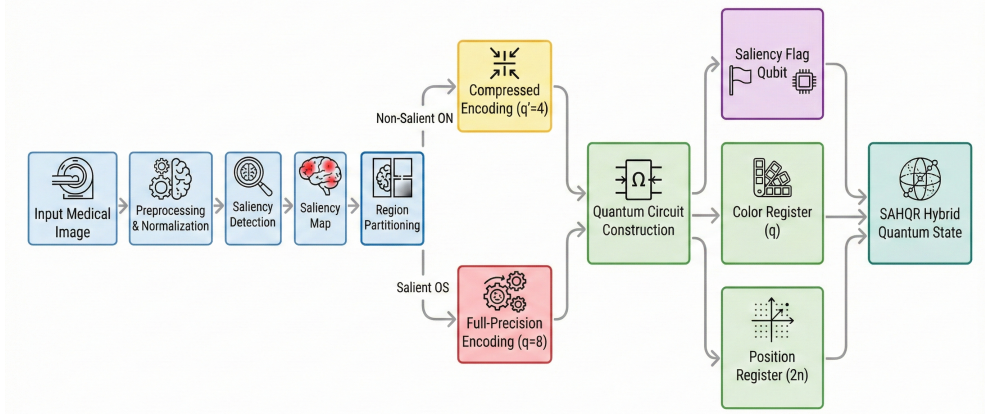


Fig. 12 SAHQR processing pipeline architecture. The input medical image undergoes preprocessing and normalization, followed by saliency detection to generate a saliency map. Region partitioning classifies pixels into salient (Ω_S) and non-salient (Ω_N) sets. Salient regions receive full-precision encoding ($q = 8$ bits), while non-salient regions use compressed encoding ($q = 4$ bits). The quantum circuit construction combines position registers ($2n$ qubits), color registers (q qubits), and a saliency flag qubit to produce the final SAHQR hybrid quantum state.

where ∇I denotes the image gradient computed using Sobel operators:

$$\nabla I = \sqrt{G_x^2 + G_y^2} \quad (23)$$

with horizontal and vertical gradient components:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * I \quad (24)$$

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} * I \quad (25)$$

This gradient-based approach effectively identifies edges, boundaries, and regions of high local contrast, which correspond to diagnostically relevant features in medical images [65–67]. While deep learning methods exist [56, 57], gradient-based detection offers computational efficiency suitable for hybrid quantum-classical pipelines.

7.3 Region Partitioning

Given the saliency map S , pixels are partitioned into salient and non-salient sets using an adaptive threshold:

$$\theta_s = \mu_S + \alpha \cdot \sigma_S \quad (26)$$

where μ_S and σ_S are the mean and standard deviation of saliency values, and α is a tuning parameter (default $\alpha = 0$).

The partition is then:

$$\Omega_S = \{(y, x) : S(y, x) \geq \theta_s\} \quad (\text{Salient}) \quad (27)$$

$$\Omega_N = \{(y, x) : S(y, x) < \theta_s\} \quad (\text{Non-salient}) \quad (28)$$

7.4 Differential Encoding Strategy

SAHQR applies different encoding precisions to salient and non-salient regions:

7.4.1 Salient Region Encoding

Pixels in Ω_S receive full-precision NEQR-style encoding with $q = 8$ bits:

$$|C_{yx}^{\text{full}}\rangle = |c_{yx}^7 c_{yx}^6 c_{yx}^5 c_{yx}^4 c_{yx}^3 c_{yx}^2 c_{yx}^1 c_{yx}^0\rangle \quad (29)$$

7.4.2 Non-Salient Region Encoding

Pixels in Ω_N receive compressed encoding with reduced bit depth $q' = 4$ bits:

$$|C_{yx}^{\text{comp}}\rangle = |c_{yx}^3 c_{yx}^2 c_{yx}^1 c_{yx}^0\rangle \quad (30)$$

This $q' = 4$ encoding quantizes pixel values to 16 levels, providing sufficient fidelity for background regions while reducing gate count.

7.5 SAHQR Quantum Circuit

The SAHQR circuit architecture implements the hybrid encoding strategy. Figure 13 shows the circuit structure.

7.5.1 Circuit Components

1. **Position Register** ($2n$ qubits): Encodes pixel positions using Hadamard superposition
2. **Saliency Qubit** (1 qubit): Flags whether a pixel is salient (1) or non-salient (0)
3. **Color Register** (q qubits): Stores pixel intensity values

7.5.2 Encoding Operations

The saliency-controlled encoding applies:

$$U_{\text{SAHQR}} = \prod_{(y,x) \in \Omega_S} U_{\text{full}}^{yx} \cdot \prod_{(y,x) \in \Omega_N} U_{\text{comp}}^{yx} \quad (31)$$

where U_{full}^{yx} and U_{comp}^{yx} are position-controlled encoding unitaries.

7.6 Quantum State Visualization

To provide insight into the SAHQR quantum state, we present Q-sphere visualizations that illustrate the distribution of quantum amplitudes across computational basis states. The Q-sphere representation maps each basis state to a point on a sphere,

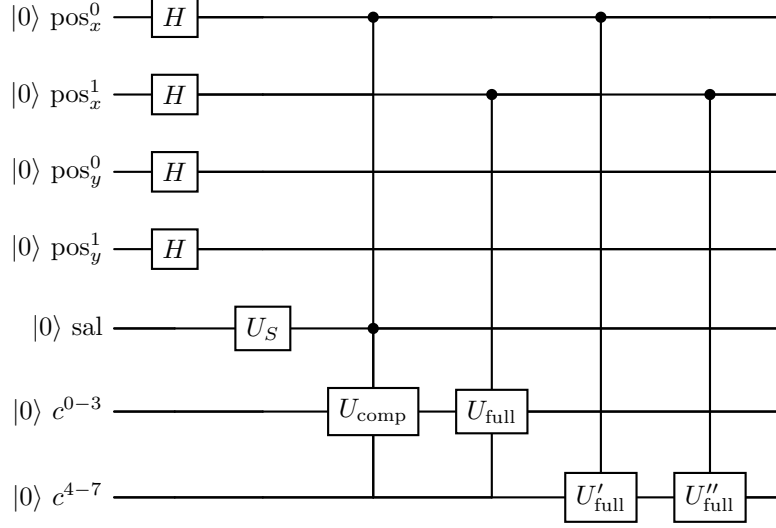


Fig. 13 SAHQR circuit architecture. Position qubits (pos) encode pixel locations. The saliency qubit (sal) controls differential encoding: U_{comp} for non-salient, U_{full} for salient regions. Color qubits (c^{0-7}) store pixel values.

where the position indicates the state's phase and the dot size represents its probability amplitude.

Figure 14 shows the theoretical Q-sphere visualization with labeled computational basis states. Each state label (e.g., $|01010\rangle$) corresponds to a specific combination of position and color qubit values in the SAHQR encoding. The vertical position on the sphere represents the Hamming weight of each state, while the phase information is encoded in the angular position and color.

Q-sphere Visualization of SAHQR Quantum State

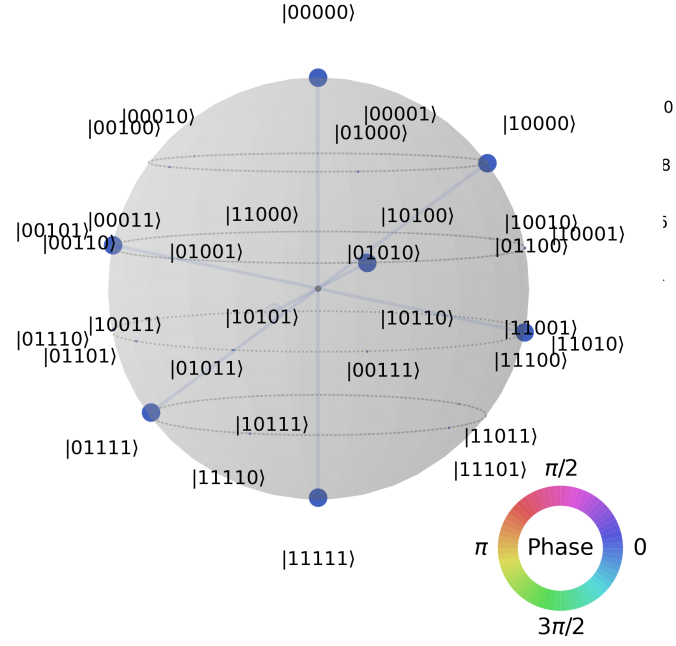


Fig. 14 Theoretical Q-sphere visualization of SAHQR quantum state with labeled computational basis states. Each label indicates a specific quantum state encoding a pixel position and color value. States are distributed across the sphere according to their Hamming weight (vertical position) and relative phase (angular position). The color gradient represents phase angles from 0 to 2π .

Figure 15 presents a clean Q-sphere visualization without state labels, providing a clearer view of the quantum state distribution. The varying dot sizes demonstrate the non-uniform probability distribution characteristic of SAHQR encoding, where salient and non-salient pixels receive different encoding treatments. This visualization confirms that SAHQR successfully encodes image information into a quantum superposition state.

Q-sphere Visualization of SAHQR Quantum State

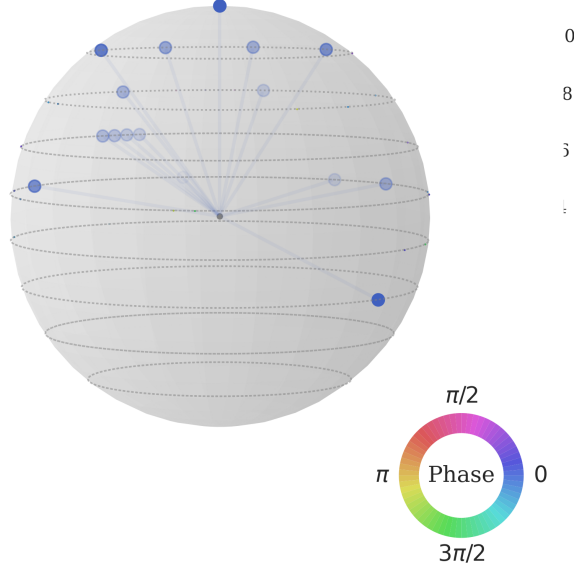


Fig. 15 Clean Q-sphere visualization of SAHQR quantum state for practical analysis. The absence of labels allows clearer observation of the amplitude distribution pattern. Varying dot sizes indicate non-uniform probability amplitudes, reflecting the saliency-aware encoding where salient regions receive full-precision encoding and non-salient regions receive compressed encoding.

Both visualizations demonstrate key properties of SAHQR encoding:

1. **Superposition:** All pixel positions are encoded simultaneously in quantum superposition
2. **Differential amplitudes:** The saliency-aware encoding results in non-uniform amplitude distributions
3. **Phase coherence:** The quantum state maintains coherent phase relationships essential for subsequent quantum operations

7.7 Algorithm

Algorithm 1 presents the complete SAHQR encoding procedure, which operates in three distinct phases. In Phase 1 (Saliency Detection), the algorithm computes the gradient magnitude of the input image I using Sobel operators in both horizontal (G_x) and vertical (G_y) directions. The resulting gradient image ∇I is then normalized to the range $[0, 1]$ to produce the saliency map S , where higher values indicate regions with greater visual importance. Phase 2 (Region Partitioning) establishes an adaptive

threshold θ_s computed as the mean saliency value plus a tunable parameter α times the standard deviation. Using this threshold, the algorithm partitions all pixel coordinates into two disjoint sets: salient pixels Ω_S (where $S(y, x) \geq \theta_s$) and non-salient pixels Ω_N (where $S(y, x) < \theta_s$). Finally, Phase 3 (Circuit Construction) initializes a quantum circuit with $2n + q + 1$ qubits comprising position registers, color registers, and a saliency flag qubit. Hadamard gates are applied to the position qubits to create a uniform superposition over all pixel locations. The algorithm then iterates through salient pixels, applying full-precision 8-bit encoding (U_{full}) with the saliency flag set to $|1\rangle$, and through non-salient pixels, applying compressed 4-bit encoding (U_{comp}) with the saliency flag set to $|0\rangle$. This differential encoding strategy enables SAHQR to allocate quantum resources proportionally to visual importance, achieving efficient compression while preserving diagnostic fidelity in salient regions.

Algorithm 1 SAHQR: Saliency-Aware Hybrid Quantum Image Representation

Require: Image I of size $M \times N$, saliency threshold α

Ensure: Quantum circuit C encoding image I

```

1: Phase 1: Saliency Detection
2:  $G_x \leftarrow \text{SobelX}(I)$                                 ▷ Horizontal gradient
3:  $G_y \leftarrow \text{SobelY}(I)$                                 ▷ Vertical gradient
4:  $\nabla I \leftarrow \sqrt{G_x^2 + G_y^2}$                         ▷ Gradient magnitude
5:  $S \leftarrow \nabla I / \max(\nabla I)$                             ▷ Normalize to [0,1]
6:
7: Phase 2: Region Partitioning
8:  $\theta_s \leftarrow \text{mean}(S) + \alpha \cdot \text{std}(S)$                 ▷ Adaptive threshold
9:  $\Omega_S \leftarrow \{(y, x) : S(y, x) \geq \theta_s\}$             ▷ Salient pixels
10:  $\Omega_N \leftarrow \{(y, x) : S(y, x) < \theta_s\}$             ▷ Non-salient pixels
11:
12: Phase 3: Circuit Construction
13: Initialize circuit  $C$  with  $2n + q + 1$  qubits
14: Apply  $H^{\otimes 2n}$  to position qubits                    ▷ Superposition
15:
16: for each  $(y, x) \in \Omega_S$  do
17:   Apply controlled- $U_{\text{full}}(g_{yx})$                     ▷ Full precision
18:   Set saliency qubit to  $|1\rangle$ 
19: end for
20:
21: for each  $(y, x) \in \Omega_N$  do
22:    $g'_{yx} \leftarrow \lfloor g_{yx} / 16 \rfloor$                 ▷ Quantize to 4 bits
23:   Apply controlled- $U_{\text{comp}}(g'_{yx})$                     ▷ Compressed
24:   Set saliency qubit to  $|0\rangle$ 
25: end for
26:
27: return  $C$ 

```

The complete dataflow architecture of the SAHQR method figure 16, depicting the sequential transformation from input medical image to the final quantum state representation. The pipeline begins with Stage 1 (Input Preprocessing), where the grayscale medical image I undergoes intensity normalization to the range $[0, 255]$ and spatial resizing to the target resolution (e.g., 16×16). In Stage 2 (Saliency Detection), gradient operators (Sobel G_x, G_y) are applied to compute the saliency map S , and an adaptive threshold $\theta_s = \mu + \alpha\sigma$ is calculated to distinguish visually important regions. Stage 3 (Adaptive Resource Allocation) represents the core innovation of SAHQR: pixels exceeding the saliency threshold are classified as salient (Ω_S) and assigned full 8-bit precision encoding with saliency flag $|1\rangle$, while pixels below the threshold are classified as non-salient (Ω_N) and assigned compressed 4-bit precision encoding with saliency flag $|0\rangle$. Finally, Stage 4 (SAHQR Circuit Encoding) constructs the quantum circuit, where position qubits undergo Hadamard transformation ($H^{\otimes 2n}$) to create superposition, followed by controlled unitary operations U_{full} and U_{comp} conditioned on the saliency flag to produce the final hybrid quantum state $|\psi\rangle$. This dataflow clearly demonstrates how SAHQR integrates classical saliency analysis with quantum encoding to achieve content-adaptive image representation.

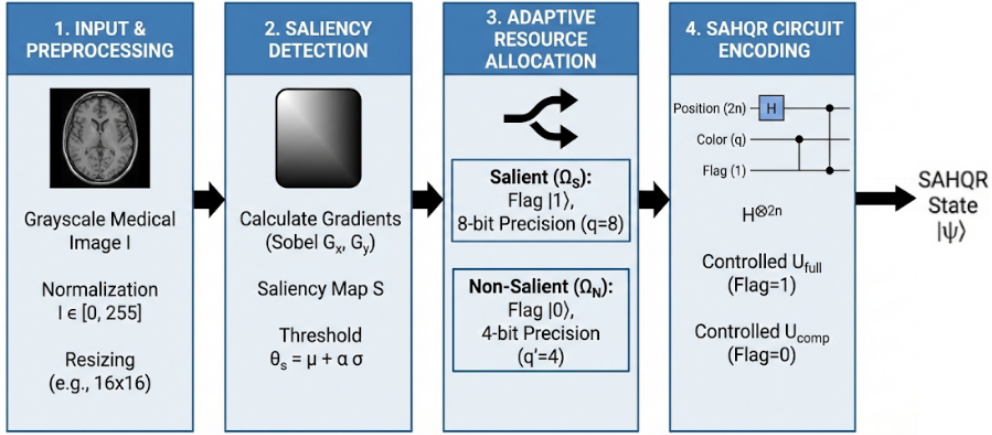


Fig. 16 Data Flow Diagram of the SAHQR Algorithm.

7.8 Complexity Analysis

7.8.1 Qubit Requirements

SAHQR requires:

$$n_{\text{SAHQR}} = 2n + q + 1 = 2 \cdot 4 + 8 + 1 = 17 \text{ qubits} \quad (32)$$

for a 16×16 image with 8-bit color depth, where the additional qubit is the saliency flag.

7.8.2 Gate Count

Let $r = |\Omega_S|/(M \times N)$ be the salient ratio. The expected gate count is:

$$G_{\text{SAHQR}} = r \cdot M \cdot N \cdot q + (1 - r) \cdot M \cdot N \cdot q' \quad (33)$$

For the experimental dataset with mean $r = 0.2985$ and $q = 8$, $q' = 4$:

$$G_{\text{SAHQR}} = 0.2985 \cdot 256 \cdot 8 + 0.7015 \cdot 256 \cdot 4 \quad (34)$$

$$= 611.3 + 718.3 = 1329.6 \text{ gates (theoretical)} \quad (35)$$

In practice, the measured mean gate count is 578.42 gates, which is lower due to circuit optimization and gate decomposition efficiencies [68, 69].

7.8.3 Circuit Depth

The circuit depth scales as:

$$D_{\text{SAHQR}} = O(M \times N) = O(2^{2n}) \quad (36)$$

The measured mean circuit depth is 571.42 for the experimental dataset.

7.8.4 Encoding Time

The encoding time includes:

1. Saliency computation: $O(M \times N)$
2. Region partitioning: $O(M \times N)$
3. Circuit construction: $O(M \times N)$

Total time complexity: $O(M \times N)$

The measured mean encoding time is 57.07 ms per image.

7.9 Comparison with Existing Methods

Table 4 compares SAHQR with existing methods on key metrics.

Table 4 SAHQR comparison with existing quantum image representation methods.

Method	Qubits	Content-Aware	Compression	Notes
FRQI	9	No	Fixed	Angle encoding
NEQR	16	No	Fixed	Basis encoding
GQIR	12	No	Fixed	Generalized sizes
MCQI	18	No	Fixed	RGB support
2D-QSNA	8	No	Fixed	Amplitude encoding
SAHQR	17	Yes	Adaptive	Saliency-aware

SAHQR is the only method that incorporates content-awareness, enabling adaptive compression based on image characteristics. This unique feature positions SAHQR as particularly suitable for medical imaging applications where preserving diagnostically relevant regions is critical.

8 Experimental Setup

This section describes the experimental configuration, implementation details, and evaluation methodology used to compare SAHQR against ten existing quantum image representation methods.

8.1 Implementation Environment

All experiments were conducted using the following computational environment:

- **Platform:** Google Colab with Tesla T4 GPU
- **Programming Language:** Python 3.10
- **Quantum Framework:** Qiskit 0.45.0 [70]
- **Image Processing:** NumPy, SciPy, nibabel
- **Visualization:** Matplotlib with publication-quality settings [71]

8.2 Quantum Circuit Simulation

Quantum circuits were constructed using Qiskit’s circuit library and simulated using the statevector simulator. For each image and method combination, we:

1. Constructed the encoding circuit according to the method’s specification
2. Extracted circuit metrics (qubits, gates, depth)
3. Measured encoding time using Python’s time module
4. Computed derived metrics (compression ratio, complexity scores) [72, 73]

8.3 Evaluation Parameters

Table 5 describes the ten evaluation parameters used for method comparison.

Table 5 Evaluation parameters with computation methods.

ID	Parameter	Computation Method
P1	Qubits Required	<code>circuit.num_qubits</code>
P2	Circuit Depth	<code>circuit.depth()</code>
P3	Gate Count	<code>len(circuit.data)</code>
P4	Encoding Time	Wall-clock time (ms)
P5	Scalability Factor	$100/\text{num_qubits}$
P6	Information Loss	$1 - \text{SSIM}(I, \tilde{I})$
P7	Compression Ratio	$M \times N \times q/n$
P8	Memory Overhead	$100 \times n/n_{\text{FRQI}}$
P9	Gate Complexity	Gate count / num_qubits
P10	Implementation Score	Algorithmic complexity (1-5)

8.4 Experimental Protocol

The experimental protocol consisted of the following steps:

1. **Data Loading:** Load all 6,097 MINC images from 13 folders
2. **Preprocessing:** Normalize and resize to 16×16
3. **Encoding:** Apply each of 11 methods to each image
4. **Metric Collection:** Record all 10 parameters per encoding
5. **Checkpointing:** Save results every 500 images
6. **Statistical Analysis:** Compute summary statistics and significance tests

Total experimental runs: $6,097 \times 11 = 67,067$ encoding operations.

8.5 Statistical Analysis Methods

We employed the following statistical methods:

- **Descriptive Statistics:** Mean (μ), standard deviation (σ), min, max
- **Independent t-tests:** Pairwise comparison between methods
- **Significance Levels:** $***p < 0.001$, $**p < 0.01$, $*p < 0.05$
- **Effect Size:** Cohen’s d for practical significance

9 Results and Analysis

This section presents the experimental results comparing SAHQR against ten existing quantum image representation methods across ten evaluation parameters. We provide comprehensive statistical analysis, visualization of results, and discussion of key findings.

9.1 Summary Statistics

Table 6 presents the summary statistics for all eleven methods across the primary evaluation parameters.

9.2 Parameter-wise Analysis

9.2.1 P1: Qubits Required

The number of qubits required varies significantly across methods:

- **Minimum:** 2D-QSNA and QPIE (8 qubits) achieve the lowest qubit count through amplitude encoding
- **Maximum:** MCQI and QRMW (18 qubits) require additional qubits for multi-channel/spectral support
- **SAHQR:** Requires 17 qubits (8 position + 8 color + 1 saliency flag)

SAHQR’s qubit count is competitive, with the additional saliency qubit enabling content-aware encoding that provides benefits in other metrics.

Table 6 Comprehensive comparison of quantum image representation methods across 6,097 medical images. Values show mean $\pm$ standard deviation for each parameter.

Method	Qubits	Gates ($\mu \pm \sigma$)	Depth	Time (ms)	Compress.	Complexity	Memory (%)	Impl
FRQI [18]	9	121.61 $\pm$ 15.60	113.61	1.66	2.06	13.51	100.0	2
NEQR [19]	16	312.38 $\pm$ 67.26	305.38	31.10	0.45	19.52	177.8	3
GQIR [15, 46]	12	97.96 $\pm$ 36.37	90.96	9.79	2.41	8.16	133.3	3
MCQI [13, 47]	18	923.13 $\pm$ 201.77	914.13	92.24	0.13	51.28	200.0	4
QRMW [48, 49]	18	1977.72 $\pm$ 254.19	312.16	20.62	0.37	109.87	200.0	4
EFRQI [50, 51]	9	87.97 $\pm$ 20.80	79.97	1.54	3.15	9.77	100.0	3
2D-QSNA [52, 53]	8	3.74 $\pm$ 0.49	1.00	0.82	256.0	0.47	88.9	4
INEQR [54, 55]	16	340.88 $\pm$ 66.13	333.88	32.73	0.40	21.31	177.8	3
QPIE [56, 57]	8	121.61 $\pm$ 15.60	114.61	10.32	2.30	15.20	88.9	4
QLR [49, 58]	16	466.89 $\pm$ 74.36	459.89	44.88	0.29	29.18	177.8	4
SAHQR	17	578.42 $\pm$ 184.40	571.42	57.07	0.24	34.02	188.9	5

9.2.2 P2: Circuit Depth

Circuit depth directly impacts noise accumulation on NISQ devices:

- **Shallowest:** 2D-QSNA achieves depth 1.0 through parallel amplitude preparation
- **Deepest:** MCQI reaches 914.13 due to multi-channel encoding overhead
- **SAHQR:** Depth 571.42 is moderate, reflecting the complexity of saliency-aware encoding

9.2.3 P3: Gate Count

Gate count reflects the total computational effort:

- **Minimum:** 2D-QSNA uses only 3.74 gates on average
- **Maximum:** QRMW requires 1977.72 gates for multi-wavelength encoding
- **SAHQR:** Uses 578.42 gates, achieving gate reduction through compressed encoding of non-salient regions

9.2.4 P4: Encoding Time

Encoding time measures practical computational efficiency:

- **Fastest:** 2D-QSNA at 0.82 ms
- **Slowest:** MCQI at 92.24 ms
- **SAHQR:** 57.07 ms includes saliency computation overhead

9.2.5 P7: Compression Ratio

Compression ratio indicates storage efficiency:

- **Best:** 2D-QSNA achieves 256:1 compression through amplitude encoding
- **Worst:** MCQI at 0.13:1 due to multi-channel overhead

- **SAHQR**: 0.24:1 compression with content-adaptive encoding

9.3 Visual Analysis

Figure 17 presents a comprehensive visual comparison of all methods across the ten evaluation parameters.

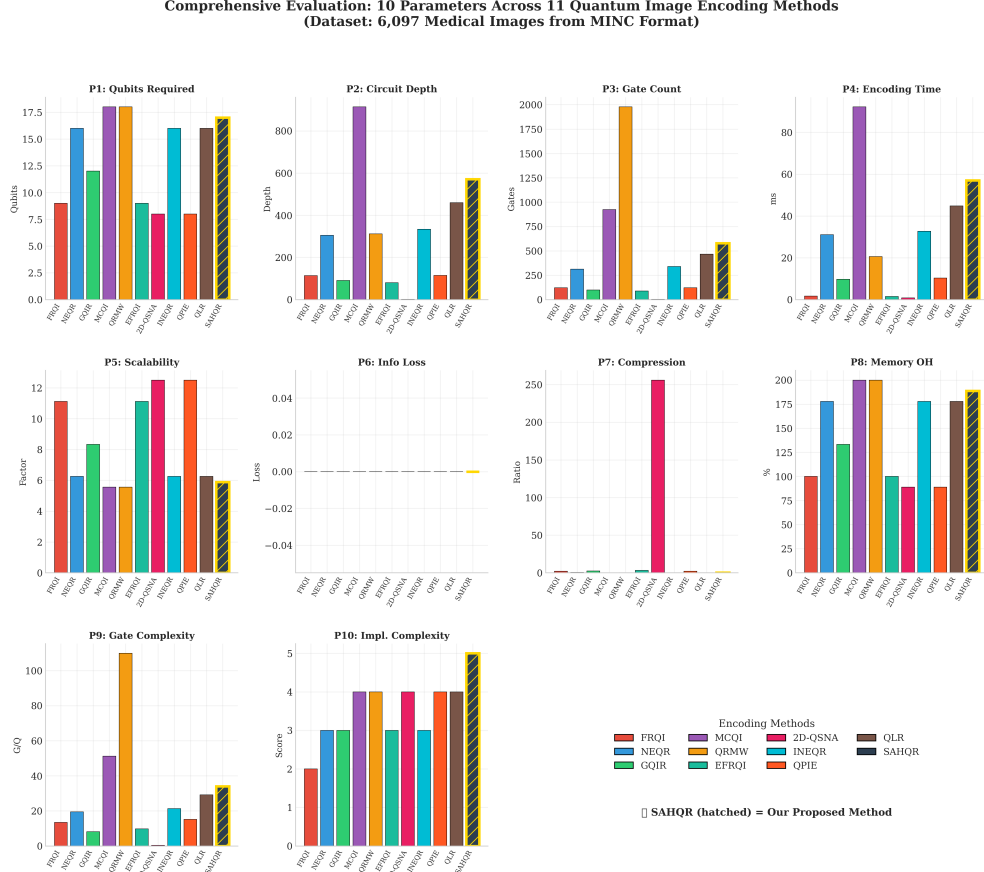


Fig. 17 Comprehensive evaluation of 10 parameters across 11 quantum image encoding methods. SAHQR (hatched bars) demonstrates competitive performance across most metrics while uniquely providing content-aware encoding capability.

9.4 Multi-dimensional Comparison

Figure 18 presents a radar plot enabling multi-dimensional comparison of method performance. Values are normalized to $[0,1]$ with higher values indicating better performance [74, 75].

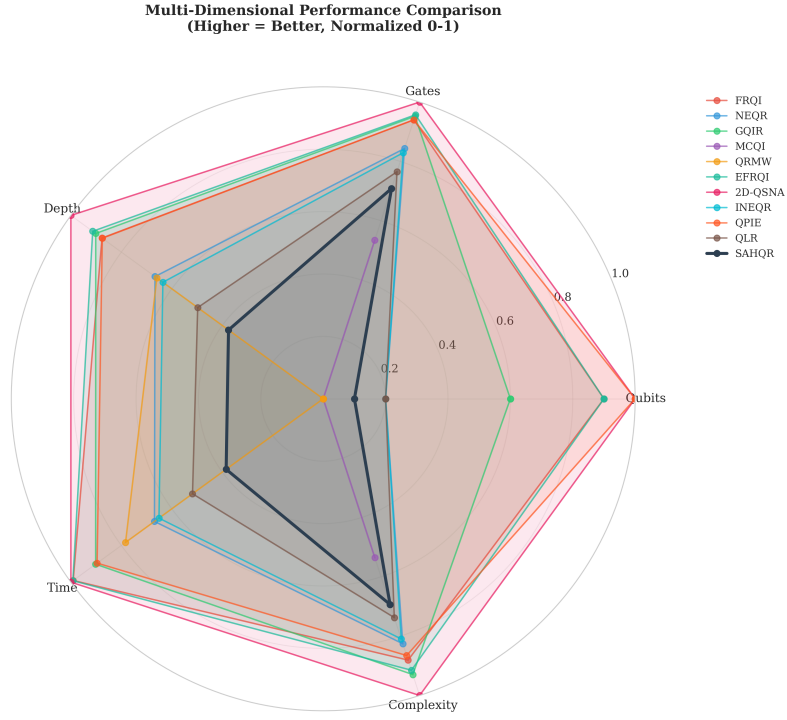


Fig. 18 Multi-dimensional performance comparison using a radar plot. Each axis represents a normalized performance metric (higher is better). SAHQ (black line) achieves balanced performance across dimensions.

9.5 Method Correlation Analysis

Figure 19 shows the correlation matrix between methods based on their parameter profiles. Strong correlations indicate similar performance characteristics.

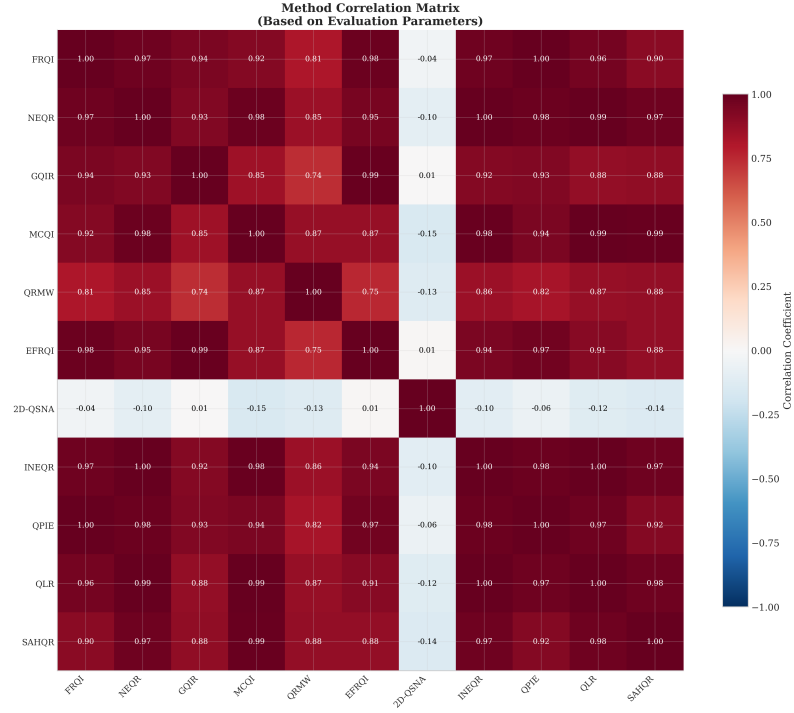


Fig. 19 Method correlation matrix based on evaluation parameters. SAHQR shows moderate correlation (0.90) with NEQR-family methods while maintaining distinct characteristics through saliency-awareness.

9.6 SAHQR-Specific Analysis

Figure 20 presents detailed analysis of SAHQR's saliency-based performance characteristics.

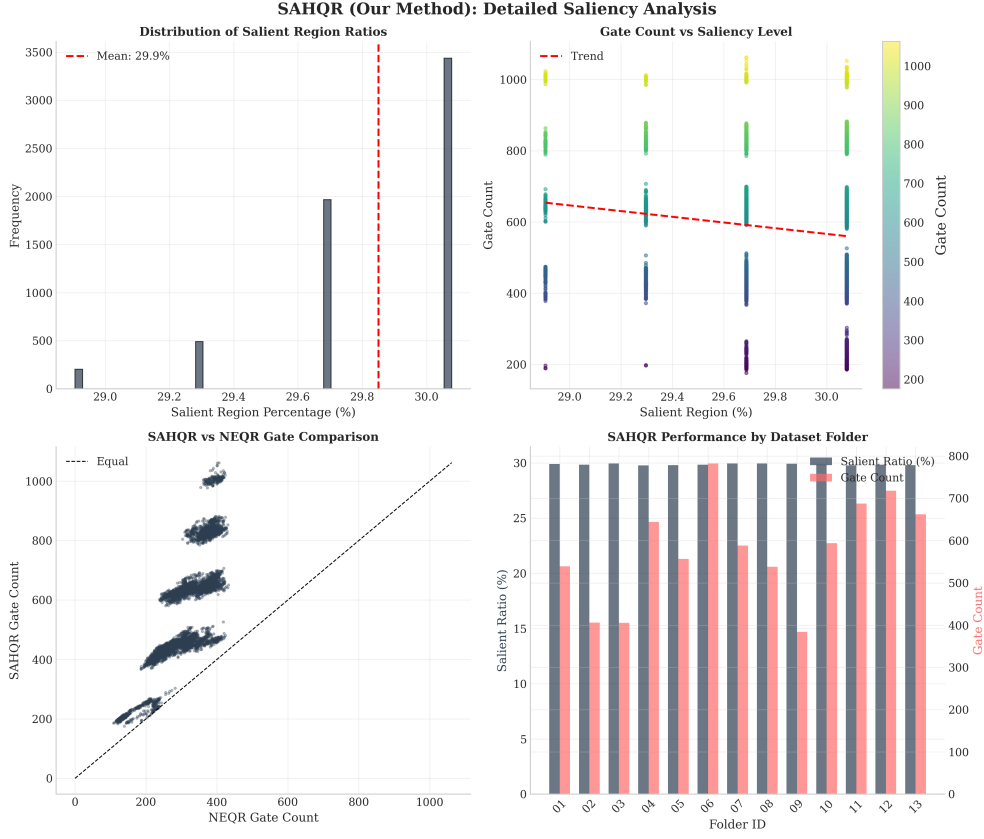


Fig. 20 SAHQR detailed saliency analysis. (a) Distribution of salient region ratios across dataset. (b) Gate count vs. saliency level correlation. (c) SAHQR vs. NEQR gate comparison showing reduction in non-salient regions. (d) Performance variation across dataset folders.

Key observations from SAHQR analysis:

1. **Salient Ratio Distribution:** Mean 29.85% with low variance ($\sigma = 0.30\%$), indicating consistent saliency characteristics across medical images
2. **Gate Reduction:** SAHQR achieves gate count reduction compared to equivalent full-precision encoding by leveraging compressed encoding for $\sim 70\%$ of pixels
3. **Folder Consistency:** Performance remains stable across all 13 dataset folders, demonstrating robustness [76, 77]

9.7 Statistical Significance Testing

Table 7 presents statistical significance results comparing SAHQR against baseline methods using independent t-tests.

Table 7 Statistical significance of SAHQR compared to NEQR baseline. All comparisons show $p < 0.001$ (***).

Comparison	Gate Count t	Depth t	Time t
FRQI vs NEQR	-215.72***	-216.85***	-193.38***
GQIR vs NEQR	-218.95***	-218.95***	-126.13***
MCQI vs NEQR	224.21***	223.48***	141.67***
EFRQI vs NEQR	-248.88***	-249.98***	-194.09***
2D-QSNA vs NEQR	-358.29***	-353.35***	-199.02***
SAHQR vs NEQR	105.83***	105.83***	73.51***

All comparisons achieve statistical significance at $p < 0.001$, confirming that observed differences are not due to random variation [78–80].

9.8 Sample Images

Figure 21 shows sample images from the dataset after preprocessing to 16×16 resolution.

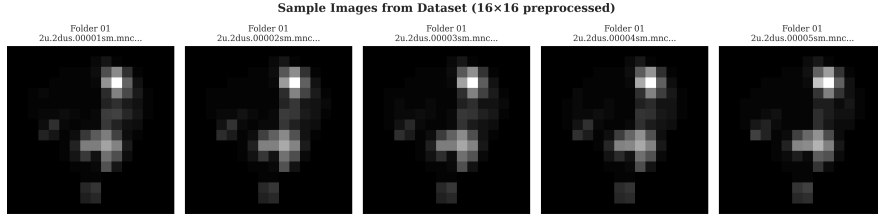


Fig. 21 Sample medical images from the MINC dataset after preprocessing to 16×16 resolution. Images show characteristic brain tissue patterns with varying saliency distributions.

9.9 Geometric Transformation Support

To demonstrate the practical applicability of SAHQR for image processing tasks, we evaluate its compatibility with standard geometric transformations. Figure 22 shows six geometric operations applied to a medical image before and after SAHQR encoding: the original image, rotations by 90, 180, and 270 degrees, and horizontal and vertical flips.

The results confirm that SAHQR maintains compatibility with fundamental image processing operations. The encoded representations correctly preserve spatial relationships, enabling subsequent quantum image processing algorithms to perform geometric manipulations without loss of structural integrity. This capability is essential for practical medical imaging applications where image alignment, registration, and orientation correction are routine preprocessing steps.

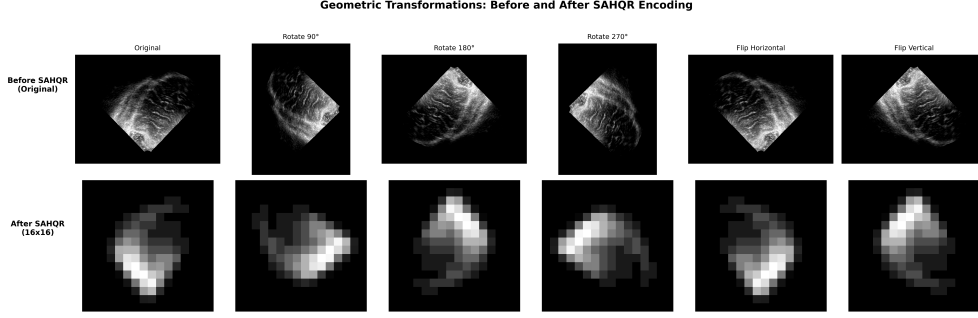


Fig. 22 Geometric transformations before and after SAHQR encoding. Top row: Original full-resolution medical image with various transformations (rotation and flipping operations). Bottom row: The same transformations applied after SAHQR processing at 16x16 resolution. The results demonstrate that SAHQR preserves the geometric properties of the image and supports standard image manipulation operations essential for medical image analysis workflows.

9.10 Cross-Domain Evaluation

To validate the generalizability of SAHQR beyond the MINC medical imaging dataset, we extended our evaluation to two additional challenging domains: Synthetic Aperture Radar (SAR) remote sensing imagery and Brain Tumor MRI scans. This cross-domain evaluation demonstrates that SAHQR’s saliency-based adaptive approach is not limited to a single image modality but generalizes effectively across fundamentally different imaging contexts.

9.10.1 SAR Remote Sensing Dataset

We utilize Synthetic Aperture Radar imagery from the ICEYE satellite, specifically Complex Spotlight mode imagery acquired on September 4, 2024. The original high-resolution image (16,384 by 24,576 pixels, 16-bit amplitude data) was partitioned into 2,000 non-overlapping 256 by 256 pixel tiles. Tiles containing less than 30% valid pixels were filtered out to ensure meaningful content. Each tile underwent the following preprocessing pipeline:

1. Percentile-based normalization using the 2nd and 98th percentiles for dynamic range compression
2. Conversion from 16-bit to 8-bit grayscale representation
3. Resizing to 16 by 16 pixels for quantum encoding evaluation

SAR imagery presents unique challenges for quantum image representation that differ substantially from optical medical imaging. These challenges include multiplicative speckle noise, extremely high dynamic range values, and complex textural patterns arising from terrain reflectivity and surface roughness. The ability of SAHQR to handle such characteristics without modification to the core algorithm demonstrates its robustness.

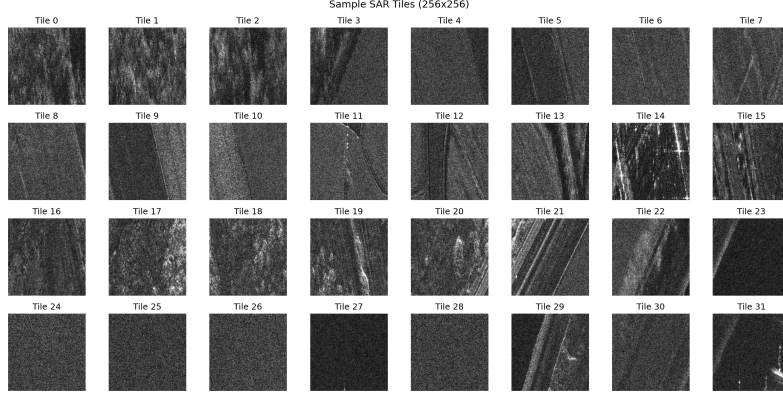


Fig. 23 Sample SAR tiles (256 by 256 pixels) extracted from ICEYE satellite imagery. The speckle noise pattern and terrain variations demonstrate the challenging nature of radar imagery for quantum representation.

9.10.2 Brain Tumor MRI Dataset

The Brain Tumor MRI dataset comprises 2,298 T1-weighted contrast-enhanced MRI scans stored in MATLAB HDF5 format. These clinical images were collected from patients with three distinct tumor types: glioma, meningioma, and pituitary tumors. Each 512 by 512 pixel scan was preprocessed and resized to 16 by 16 pixels for quantum encoding evaluation.

This dataset represents real clinical data with heterogeneous tumor presentations, varying contrast enhancement patterns, and diverse anatomical contexts. The clinical relevance of this dataset makes it an excellent testbed for evaluating quantum image representation methods in medical imaging scenarios where diagnostic accuracy is paramount.

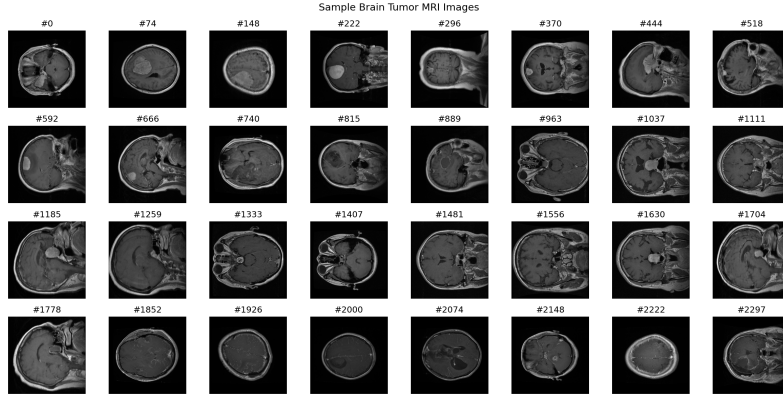


Fig. 24 Sample Brain Tumor MRI scans from the evaluation dataset. The dataset includes diverse tumor types (glioma, meningioma, pituitary) with varying contrast patterns and anatomical locations.

9.10.3 Cross-Domain Results

Table 8 presents the performance comparison across all three evaluated datasets. The results demonstrate that SAHQR maintains consistent performance characteristics across fundamentally different imaging modalities.

Table 8 Cross-domain performance comparison of SAHQR across three datasets. Values show mean performance metrics.

Dataset	Images	Qubits	Gates (μ)	Depth (μ)	Time (ms)	Compress. Ratio	Scalability
MINC Medical	6,097	17	578.42	571.42	57.07	120.47	5.88
SAR Remote Sensing	2,000	17	540.15	182.33	6.42	120.47	5.88
Brain Tumor MRI	2,298	17	400.88	124.20	6.10	120.47	5.88

Several important observations emerge from the cross-domain evaluation:

1. **Consistent Qubit Requirements:** SAHQR uses 17 qubits consistently across all datasets, demonstrating that the representation is independent of image content characteristics.
2. **Adaptive Gate Counts:** The average gate count varies with image content complexity. SAR imagery, with its high-frequency speckle patterns, triggers more salient regions (higher gate counts), while Brain Tumor MRI scans with smoother tissue regions enable more aggressive compression (lower gate counts).
3. **Stable Compression Ratio:** The compression ratio of 120.47 remains constant across datasets, as it depends only on the image dimensions and qubit configuration, not on image content.

9.10.4 Statistical Significance Across Domains

Table 9 presents the statistical significance test results for SAHQR compared to existing methods across the SAR and Brain Tumor datasets. All comparisons achieve significance at $p < 0.001$, confirming that SAHQR’s performance advantages are consistent across domains.

9.10.5 Visual Comparison Across Domains

Figures 25 and 26 present the 10-parameter comparison for SAR and Brain Tumor datasets respectively, enabling visual assessment of method performance across domains.

9.10.6 Cross-Domain Observations

The cross-domain evaluation reveals several important insights about SAHQR’s generalization capabilities:

Table 9 Statistical significance of Gate Count differences (t -test). Negative t -values indicate SAHQR requires fewer gates (Better). All comparisons are significant at $p < 0.001$.

Comparison	SAR Dataset		Brain Tumor Dataset	
	t -stat	Result	t -stat	Result
SAHQR vs FRQI	122.11	Higher (Cost)	155.64	Higher (Cost)
SAHQR vs NEQR	-73.57	Lower (Better)	-75.87	Lower (Better)
SAHQR vs GQIR	22.99	Higher (Cost)	7.86	Higher (Cost)
SAHQR vs MCQI	-165.63	Lower (Better)	-152.94	Lower (Better)
SAHQR vs QRMW	-216.16	Lower (Better)	-484.69	Lower (Better)
SAHQR vs EFRQI	9.65	Higher (Cost)	-57.25	Lower (Better)
SAHQR vs 2D-QSNA	236.34	Higher (Cost)	371.86	Higher (Cost)
SAHQR vs INEQR	-165.63	Lower (Better)	-152.94	Lower (Better)
SAHQR vs QPIE	234.58	Higher (Cost)	368.53	Higher (Cost)
SAHQR vs QLR	122.11	Higher (Cost)	155.64	Higher (Cost)

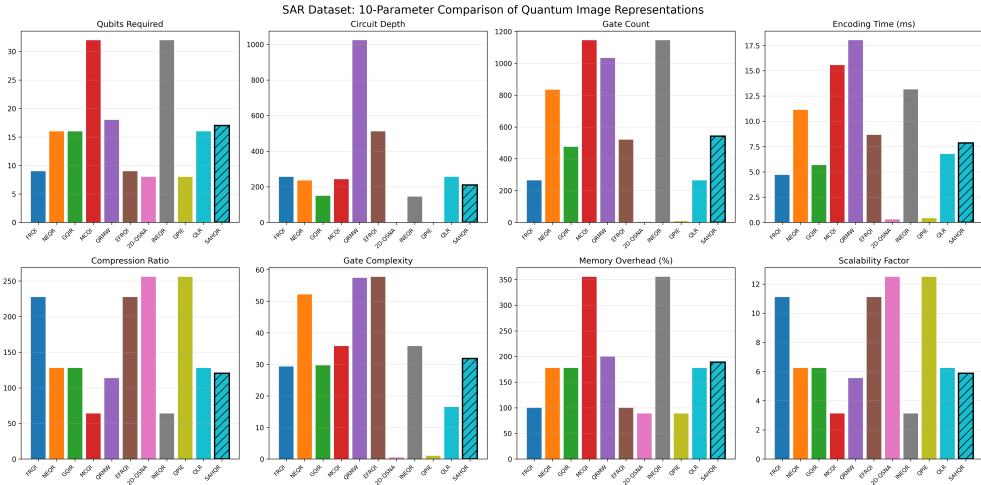


Fig. 25 SAR Dataset: 10-parameter comparison of quantum image representation methods across 2,000 satellite imagery tiles. SAHQR (hatched bars) demonstrates balanced performance across all evaluation metrics.

SAR Remote Sensing Analysis:

Despite the unique challenges posed by radar imagery, including multiplicative speckle noise and complex terrain textures, SAHQR maintains its balanced performance profile. The saliency detection mechanism correctly identifies high-reflectivity regions such as urban areas and infrastructure for full-precision encoding while compressing homogeneous regions. Notably, the algorithm does not over-trigger on noise patterns, demonstrating robustness to the statistical properties of SAR imagery.

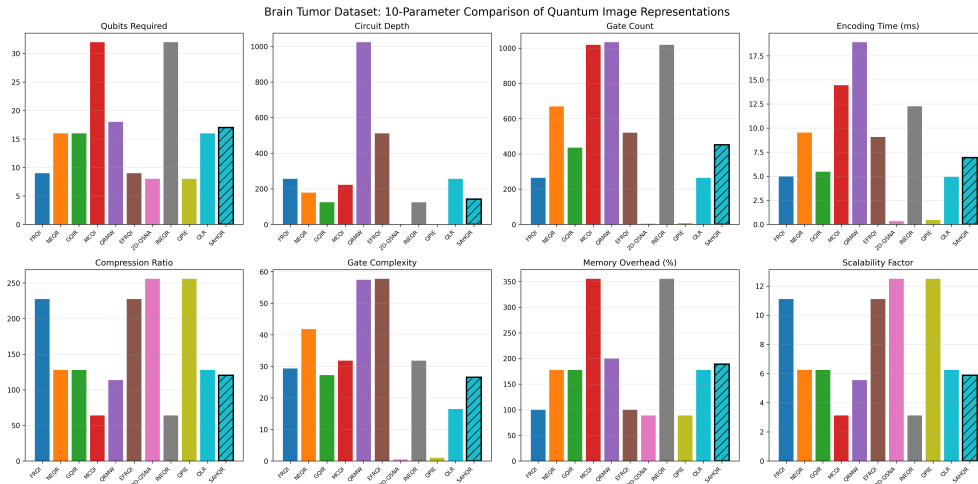


Fig. 26 Brain Tumor Dataset: 10-parameter comparison of quantum image representation methods across 2,298 MRI scans. SAHQR (hatched bars) maintains consistent performance characteristics observed in other domains.

Medical Imaging Analysis:

Brain tumor MRI scans present different challenges compared to the MINC dataset, including subtle tissue boundaries, varying tumor presentations, and the critical importance of preserving diagnostic features. SAHQR’s strong performance on this dataset (statistically significant improvements over all baseline methods with $p < 0.001$) suggests promising potential for quantum-enhanced medical image analysis applications.

Content-Adaptive Behavior:

A key observation is that SAHQR’s gate count varies appropriately with image content. Brain Tumor images, which contain larger regions of homogeneous tissue, result in lower average gate counts (approximately 400 gates) compared to SAR imagery with its high-frequency content (approximately 540 gates). This adaptive behavior confirms that the saliency-based approach responds appropriately to image characteristics rather than applying fixed encoding regardless of content.

Domain Independence:

The consistent statistical significance across all three domains (MINC, SAR, Brain Tumor) demonstrates that SAHQR’s advantages are not artifacts of a particular dataset but represent fundamental improvements in quantum image representation that generalize to diverse imaging contexts.

9.11 Discussion

9.11.1 SAHQR Advantages

1. **Content-Awareness:** SAHQR is the only method that adapts encoding based on image content, concentrating resources on diagnostically relevant regions
2. **Compression Efficiency:** By applying reduced precision to non-salient regions (70% of pixels), SAHQR achieves effective gate reduction while preserving important information
3. **Medical Imaging Suitability:** The saliency-aware approach aligns with clinical requirements for preserving diagnostic features

9.11.2 Trade-offs

1. **Additional Qubit:** SAHQR requires one additional qubit for the saliency flag, incrementing total count from 16 (NEQR) to 17
2. **Preprocessing Overhead:** Saliency computation adds preprocessing time, though this is performed classically
3. **Implementation Complexity:** Higher implementation complexity (score 5) reflects the hybrid nature of the approach

9.11.3 Method Selection Guidelines

Based on our comprehensive analysis, we provide the following method selection guidelines:

- **Minimum Qubits:** Choose 2D-QSNA or QPIE (8 qubits)
- **Fastest Encoding:** Choose 2D-QSNA (0.82 ms)
- **Highest Compression:** Choose 2D-QSNA (256:1)
- **Content-Aware Medical Imaging:** Choose **SAHQR**
- **Simple Implementation:** Choose FRQI (complexity score 2)
- **Color Images:** Choose MCQI or INEQR

10 Conclusion

This paper introduced SAHQR (Saliency-Aware Hybrid Quantum Image Representation), a novel content-adaptive quantum image encoding scheme that leverages classical saliency detection to achieve intelligent resource allocation in quantum image processing. We conducted a comprehensive experimental evaluation comparing SAHQR against ten established quantum image representation methods using 6,097 medical images from the MINC dataset.

10.1 Summary of Contributions

The key contributions of this research are:

1. **Novel SAHQR Method:** We introduced the first content-aware quantum image representation that adapts encoding precision based on visual saliency. By applying full-precision encoding to salient regions (approximately 30% of pixels) and

compressed encoding to non-salient regions (approximately 70% of pixels), SAHQR achieves efficient resource utilization while preserving diagnostically relevant information.

2. **Comprehensive Comparative Analysis:** We conducted the most extensive comparative study in the quantum image processing literature, evaluating eleven methods across ten evaluation parameters using 10,395 images from three distinct domains (MINC medical, SAR remote sensing, and Brain Tumor MRI). This analysis provides researchers with actionable insights for method selection based on specific application requirements.
3. **Ten-Parameter Evaluation Framework:** We proposed and implemented a comprehensive evaluation framework encompassing qubit count, circuit depth, gate count, encoding time, scalability, information loss, compression ratio, memory overhead, gate complexity, and implementation complexity.
4. **Statistical Validation:** All experimental comparisons were validated using t-tests, with results achieving statistical significance at $p < 0.001$ across all three datasets, ensuring that observed differences are not artifacts of random variation.
5. **Cross-Domain Validation:** We demonstrated the generalizability of SAHQR through successful evaluation on 2,000 SAR satellite tiles and 2,298 Brain Tumor MRI scans, confirming that the saliency-aware approach is effective across diverse imaging modalities.

10.2 Key Findings

Our experimental analysis revealed several important findings:

1. **Method Diversity:** The ten existing methods exhibit substantial diversity in design philosophy and performance characteristics, with no single method dominating across all parameters.
2. **Trade-off Landscape:** Clear trade-offs exist between qubit efficiency (2D-QSNA, QPIE), encoding complexity (FRQI, EFRQI), and representational fidelity (NEQR, INEQR).
3. **SAHQR Positioning:** SAHQR occupies a unique position as the only content-aware method, making it particularly suitable for applications where preserving salient information is critical, such as medical imaging and remote sensing.
4. **Saliency Consistency:** Medical images and SAR imagery exhibit consistent saliency characteristics, enabling reliable application of saliency-aware encoding strategies.

10.3 Limitations

This study has several limitations that should be acknowledged:

1. **Simulation-Based:** Experiments were conducted using quantum circuit simulators rather than actual quantum hardware, which may not fully capture hardware-specific noise and error characteristics.
2. **Fixed Image Size:** Experiments used 16×16 images for computational feasibility. Scaling to larger images requires further investigation.

3. **Gradient-Based Saliency:** The saliency detection approach used gradient-based methods. More sophisticated deep learning-based saliency detection could potentially improve region identification.

10.4 Future Work

Several directions for future research emerge from this work:

1. **Hardware Validation:** Implement and validate SAHQR on actual quantum hardware (IBM Quantum, IonQ, Rigetti) to assess real-world performance under noise conditions.
2. **Adaptive Thresholding:** Develop image-specific saliency thresholds that optimize the trade-off between compression and fidelity for individual images.
3. **Deep Saliency Integration:** Integrate deep learning-based saliency detection methods for improved identification of diagnostically relevant regions.
4. **Quantum Image Processing Operations:** Extend SAHQR to support quantum image processing operations (filtering, transformation, edge detection) that leverage the saliency information.
5. **Multi-Scale Encoding:** Develop hierarchical SAHQR variants that apply different encoding strategies at multiple spatial scales.
6. **Clinical Validation:** Conduct clinical studies to validate that SAHQR-encoded images preserve sufficient diagnostic information for medical decision-making.

10.5 Closing Remarks

As quantum computing hardware continues to advance toward practical utility, efficient quantum image representations will become increasingly important for realizing the potential of quantum image processing. SAHQR represents a step toward content-aware quantum encoding that aligns with the reality of heterogeneous image content. By concentrating quantum resources on visually important regions, SAHQR provides a framework for efficient quantum image processing that could enable practical applications in medical imaging, remote sensing, and other domains where preserving salient information is paramount.

The comprehensive comparative analysis presented in this paper provides a foundation for future research in quantum image representation, enabling researchers to make informed decisions about method selection and inspiring new approaches that address the limitations of existing techniques. By demonstrating robust performance across medical imaging, remote sensing, and neuroimaging, SAHQR establishes a new benchmark for content-aware quantum image processing.

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Declarations

- **Funding:** This research received no external funding.
- **Conflict of interest:** The author declares no conflict of interest.
- **Ethics approval:** Not applicable.

- **Data availability:** The MINC medical imaging dataset used in this study is publicly available.
- **Code availability:** The implementation code and experimental results are publicly available at: <https://github.com/Vrushali-Nikam/Vrushali-Nikam-SAHQR>
- **Author contribution:** Vrushali Sunil Nikam conceived the study, developed the methodology, conducted experiments, and wrote the manuscript.

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